
Scribble-based Image Segmentation with Convexity Constraints

Lea Dielmann^a

`lea.dielmann@student.uni-siegen.de`

Andreas Lizenberger^a

`andreas.lizenberger@student.uni-siegen.de`

^aDepartment of Electrical Engineering and Computer Science
University of Siegen
Hölderlinstraße 3, 57076 Siegen

Abstract

We present an approach to segment convex objects with neural networks and user-generated scribbles. Our approach avoids collecting massive amounts of training data and runs on low-end machines like laptops. We compose two simple neural networks that produce a mathematically convex segmentation mask. The Intersection-over-Union score ranges around 90% for the best results, but can drop as low as 50% depending on the provided scribbles and images.

1 Introduction

A common task in the field of image segmentation is to process digital images to create pixel-based labels. In the easiest case the labels differentiate between foreground and background. This is useful for image editing software to create selections of specific objects.

A difficult challenge when training neural networks is the acquisition of suitable training data. It is often not possible or impractical to access vast amounts of labeled examples. A scribble-based approach creates training data by asking the user to scribbles on the image. Scribbles are simple lines or areas that denote which part of the image is background and which part of the image is foreground. Scribbles can be viewed as an incomplete segmentation. A neural network can be trained on scribbles to create predictions for all pixels. This creates the complete segmentation. Scribble-based segmentation with neural networks has been investigated by many different authors [3] [8] [5].

Convex objects have clear boundaries, imagine a circle or square. Neural networks that do not respect these boundaries create undesired fragments. We propose a composition of two simple neural networks that takes advantage of these convexity constraints resulting in a clean convex shape. First a fully convolutional network creates a complete segmentation from the given scribbles [7]. Then an input convex neural network computes the convex shape [1].

2 Related Work

Image segmentation plays an important role in detection of health related problems. Many objects have a convex shape and must be segmented precisely. There has been an approach with respect to convexity but it relies on large models and large training data [6].

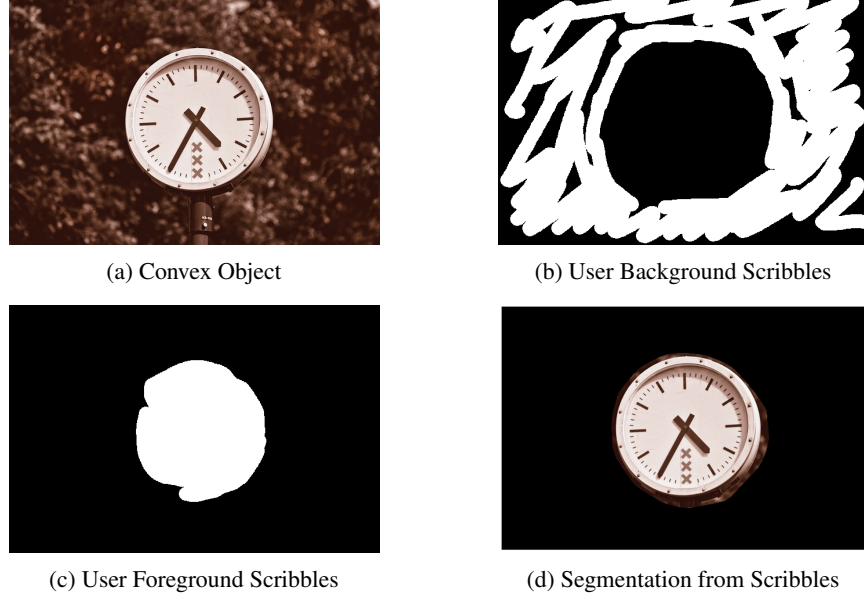


Figure 1: Segmentation process

The Segment Anything project trains a foundation model to create segmentation masks from textual prompts. They use over 11 million images and 1 billion segmentation masks for training. Zero-shot transfer enables the neural network to solve tasks it was not specifically trained for. Unfortunately, suggesting a prompt that restricts the output to a convex shape does not mathematically guarantee that the neural network will produce said convex shape. Additionally, the neural network might fail for terminology it has not been trained on, e.g. a medical term that it has never seen. This could require more extensive descriptions of the object that is supposed to be segmented and can introduce more ambiguity to the prompt [4].

3 Methodology

Let us outline the general setup. There are 2 neural networks:

- Network 1 is a fully convolutional network that creates an initial segmentation [7].
- Network 2 is a input convex neural network that creates the final convex segmentation [1].

Our pipeline follows the following steps:

1. Let the user create scribbles for the foreground.
2. Let the user create scribbles for the background.
3. Extract training data of the form (x, y, r, g, b) for each scribbled pixel. The labels are provided in step 1 and step 2.
4. Train network 1 to classify 3×3 patches of each pixel in the training data.
5. Create a segmentation mask with network 1.
6. Use that mask to train network 2.
7. Create a final convex segmentation with network 2.

The scribbles are an incomplete and rough segmentation provided by the user. Network 1 improves on the given scribbles and attempts to create a segmentation on the entire image. Network 2 can improve the result further because the foreground object must be convex. Why are two separate neural networks required?

Let img be the input image. A neural network for segmentation can be viewed as a function f that maps each pixel $img[y, x] := (x, y, r, g, b)$ to $[0, 1]$. The output expresses the likelihood that the

pixel is foreground or background. We define foreground as 0 and background as 1. The selection is not arbitrary but rather a consequence of the convexity constraints. Let us provide a definition for convex functions [2].

Theorem 1 *A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if $\forall z_1, z_2 \in \mathbb{R}^n$ and $0 \leq \alpha \leq 1$ it holds:*

$$f((1 - \alpha)z_1 + \alpha z_2) \leq (1 - \alpha)f(z_1) + \alpha f(z_2)$$

The function f must stay underneath the direct line that connects two points $(z_1, f(z_1)), (z_2, f(z_2))$. This means the output of the network must be close to 0 around the convex foreground and close to 1 everywhere else.

Unfortunately, we cannot use a single neural network f of the form $f: \mathbb{R}^5 \rightarrow \mathbb{R}$. As stated earlier, we require a direct line between two points. In an image a direct line is an interpolation of the spatial components and only those components. We must exclude the color components.

Example 2 *Let $img[y_1, x_1]$ and $img[y_2, x_2]$ be two pixels in the image. Select $z_1 = img[y_1, x_1]$ and $z_2 = img[y_2, x_2]$. A direct line that connects $(z_1, f(z_1)), (z_2, f(z_2))$ is given by the interpolation of the two points. We find that interpolating the two points also interpolates the color.*

We conclude that network 2 must have the form $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ mapping (x, y) coordinates to $[0, 1]$. Network 1 has the form $f: (\mathbb{R}^5)^9 \rightarrow \mathbb{R}$ mapping each 3×3 patch to $[0, 1]$. We use network 1 to compensate for the lost color components. Network 2 then attempts to create a similar segmentation by fitting a convex area into the output of network 1.

The architecture of network 1 has been chosen arbitrarily and can be replaced with any other architecture that segments images. Convolutional neural networks are a straightforward approach. They are easy to train, can be kept fairly small and can be applied in a simple manner on images of any size. The training is done on 3×3 patches; around the edges we added padding of length 1 and duplicated the edge information. Inference can then be performed on the padded image as input. Fully convolutional networks consist of convolutional layers, pooling layers and activation functions, therefore can be applied on inputs of any size in the spatial dimensions.

Let us discuss how input convex neural networks work. They consist of:

- fully connected layers $W_i^{(z)}$
- skip connections $W_i^{(y)}$
- and activation functions g_i .

The following theorem describes the architecture and how to achieve the convexity property [1].

Theorem 3 *A neural network $f(y; \theta) = z_k$ with layers*

$$z_1 = g_0 (W_0^{(y)}(y, \theta_0^{(y)}))$$

and

$$z_{i+1} = g_i (W_i^{(z)}(z_i, \theta_i^{(z)}) + W_i^{(y)}(y, \theta_i^{(y)}))$$

and weights $\theta = \{\theta_0^{(y)}, \theta_1^{(y)}, \dots, \theta_{k-1}^{(y)}\} \cup \{\theta_1^{(z)}, \theta_2^{(z)}, \dots, \theta_{k-1}^{(z)}\}$ is convex if

- $\theta_i^{(z)}$ are non-negative, $1 \leq i \leq k - 1$
- and all activation functions g_i are convex and non-decreasing.

We can achieve the first condition by clamping the weights after each backward pass during training. Clamping means replacing all negative values with 0. In practice the remaining weights in the neural network must adjust to compensate for the given restriction. For the second condition we can pick the ReLU as activation function for all g_i . The theorem is a consequence of the next two theorems [2].

Theorem 4 *Let g, h be convex functions and $0 \leq \alpha, \beta \in \mathbb{R}$ then $\alpha g + \beta h$ is a convex function.*

Theorem 5 *Let $g: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $h: \mathbb{R}^m \rightarrow \mathbb{R}$ be convex functions. If h is non-decreasing then the composition $h \circ g$ is a convex function.*

4 Experiments

In this section we give an overview how well our approach works on different images. We investigate whether our setup can segment images in a real world scenario and what kind of images remain a challenge. We select a variety of images, some of which do not depict convex objects in a strictly mathematical sense, to achieve a realistic evaluation. We manually create optimal segmentations for each image using photo editing software and compare those segmentations to the output of the neural networks. We use Intersection-over-Union (IoU) to measure the quality of the results. The shape of the results are convex areas and we are interested in the overlap of these areas. The scribbles for the training are kept intentionally inaccurate and rough to simulate a realistic application of our pipeline.

A summary of our results is shown in Figure 2. The boxplot shows the median of 20 experiments per image (vertical line within box), the interquartile range (box), maximum and minimum (whiskers), outliers (diamonds). Outliers are outside of 1.5 times the interquartile range.

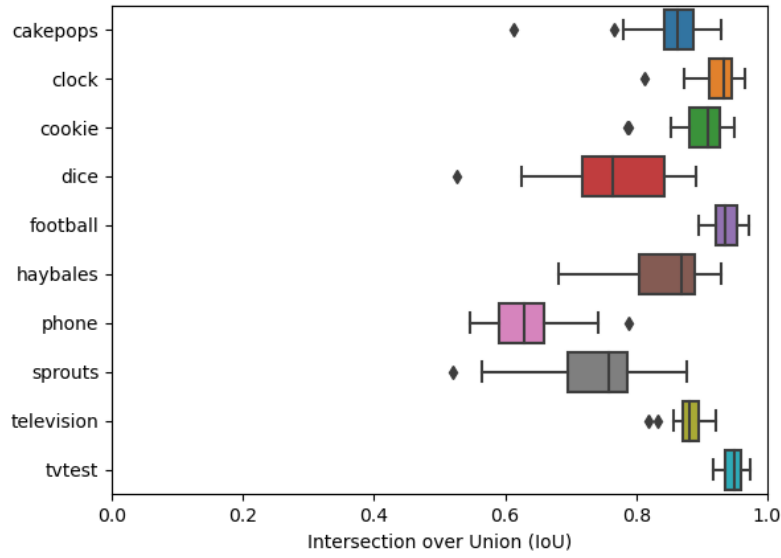


Figure 2: Performance on 10 Images

The segmentation masks vary in their respective IoU-Scores. The neural networks perform well on most round objects (cakepops, clock, cookie, football, sprouts, tvtest) but struggle with most angular objects (dice, haybales, phone, television). We assume that this effect occurs because we use a round brush to create the scribbles. Corners of angular objects are neglected. There could be a problem with the expressiveness of the networks since we optimized the hyperparameters on an image with a round object. “Sprouts” seem to have poor results due to challenging boundaries as opposed to “television” thanks to its clear boundaries.

It appears that the neural networks favor the spatial coordinates of the scribbles over the color information in the image. In our current approach we train the first network with all provided scribbles. When the object is much smaller than the background then the object has proportionally less representation in the training data. The neural networks might be biased towards the background scribbles. This could be another reason why angular objects are not properly segmented.

The output of the second network strongly relies on the output of the first network. If the first network provides poor results then the error propagates into the second network. The second network just fits the most optimal convex shape into the given segmentation mask and does not correct the results of the first network. This problem occurs most prominently in the “phone” image.

Our networks must be retrained on each image. We use a standard training approach because it is difficult to find hyperparameters that work for any image. The number of layers and hidden neurons of the networks have been determined by optimizing for a specific test image depicting “apples”. We

achieved a mean IoU of 0.9276, low IoU of 0.8115 and top IoU of 0.9690 showing that our approach has the potential to deliver good results. The fully convolutional network has the following setup:

- 1 convolutional layer of size 3×3 and 20 feature channels
- 4 convolutional layers of size 1×1 and 40, 40, 20, 1 feature channels
- all convolutional layers use a stride of 1
- after each layer follows a ReLU, and a sigmoid function after the last layer

The input convex neural network has the following form:

- 6 fully connected layer of output sizes 20, 40, 40, 40, 20, 1
- 5 skip connections of output sizes 40, 40, 40, 20, 1
- the sum of skip connection and fully connected layer is passed through a ReLU
- the final output is passed through a sigmoid function

5 Conclusion

We have demonstrated that the composition of two simple neural networks can effectively segment convex objects based on user scribbles. The first network is a fully convolutional network that creates an initial segmentation of the image. The second network is a input convex neural network that fits the convex shape into that initial segmentation.

The networks perform best for round objects and are open for improvement for other convex shapes e.g. angular convex objects.

There could be improvements made to the training, e.g. balancing the number of foreground and background scribbles. Not all scribbled pixels are equally relevant for training. Pruning some of the scribbles could reduce the duration of training and improve generalization. This could help the network to rely less on the spatial data and put more weight on the color.

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